



(b) Metsulfuron-methyl acts by inhibiting an enzyme called acetolactate synthetase, which is needed for the daisies and red poppies to synthesise three amino acids essential for growth. The specific mutation carried by the red poppies of biotype X occurred within the gene coding for this enzyme. The mutation changed amino acid 197 of acetolactate synthetase from proline to leucine.

(i) Suggest the effect of this mutation on the structure and activity of the acetolactate synthetase enzyme of biotype X red poppies.

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(ii) Suggest the effect of this mutation on the biotype X red poppies in the presence of metsulfuron-methyl.

..... [1]

(iii) With reference to Fig. 5.1, predict **and** explain the effect of biotype X red poppies on the relative proportions of weeds and wheat in the area of wheat fields in 2003 compared to 1998.

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- (c) Suggest how stopping the use of the herbicide metsulfuron-methyl and replacing it with a herbicide that inhibits a different target enzyme in weeds would affect the abundance of red poppies of biotype X.

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[Total: 14]